

**Demonstrate the working of the decision tree based ID3 algorithm.
Use an appropriate data set for building the decision tree and apply
this knowledge to classify a new sample.**

Aim

To implement and demonstrate the **Decision Tree based ID3 algorithm** using an appropriate dataset and classify a new sample using Python in Google Colab.

Apparatus / Requirements

- Google Colab

Procedure

1. Open Google Colab and create a new notebook.
2. Import the required Python libraries (pandas, scikit-learn, and matplotlib).
3. Create or load the Play Tennis dataset.
4. Convert the categorical attributes into numerical values using **Label Encoding**.
5. Separate the dataset into feature variables (X) and the target variable (Y).
6. Train the Decision Tree model using the **ID3 algorithm** by setting the criterion as **entropy**.
7. Visualize the generated Decision Tree.
8. Provide a new sample to the trained model.
9. Predict and display the class label (Play Tennis = Yes/No) for the new sample.
10. Analyze the prediction and verify the result.

code

```
# Import libraries
```

```
import pandas as pd
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt
```

```
data = {
    'Outlook': ['Sunny','Sunny','Overcast','Rain','Rain','Rain',
               'Overcast','Sunny','Sunny','Rain','Sunny',
               'Overcast','Overcast','Rain'],

    'Temperature': ['Hot','Hot','Hot','Mild','Cool','Cool',
                   'Cool','Mild','Cool','Mild','Mild',
                   'Mild','Hot','Mild'],

    'Humidity': ['High','High','High','High','Normal','Normal',
                'Normal','High','Normal','Normal','Normal',
                'High','Normal','High'],

    'Wind': ['Weak','Strong','Weak','Weak','Weak','Strong',
            'Strong','Weak','Weak','Weak','Strong',
            'Strong','Weak','Strong'],

    'PlayTennis': ['No','No','Yes','Yes','Yes','No',
                  'Yes','No','Yes','Yes','Yes',
                  'Yes','Yes','No']
}
```

```
df = pd.DataFrame(data)
```

```
print("Dataset:\n")
```

```
print(df)
```

```
le = LabelEncoder()
```

```
df_encoded = df.copy()
```

```
for column in df.columns:
```

```
    df_encoded[column] = le.fit_transform(df[column])
```

```
print("\nEncoded Dataset:\n")
```

```
print(df_encoded)
```

```
X = df_encoded.drop('PlayTennis', axis=1)
```

```
y = df_encoded['PlayTennis']
```

```
model = DecisionTreeClassifier(criterion='entropy')
```

```
model.fit(X, y)
```

```
print("\nDecision Tree Model Trained Successfully!")
```

```
plt.figure(figsize=(12,8))
```

```
plot_tree(model,  
          feature_names=X.columns,  
          class_names=['No','Yes'],  
          filled=True)  
  
plt.title("Decision Tree using ID3 Algorithm")  
plt.show()
```

```
# New Sample:  
# Outlook = Sunny  
# Temperature = Cool  
# Humidity = High  
# Wind = Strong
```

```
sample = pd.DataFrame({  
    'Outlook':['Sunny'],  
    'Temperature':['Cool'],  
    'Humidity':['High'],  
    'Wind':['Strong']  
})
```

```
# Encode sample manually according to training data
```

```
sample_encoded = pd.DataFrame()
```

```
for column in sample.columns:
```

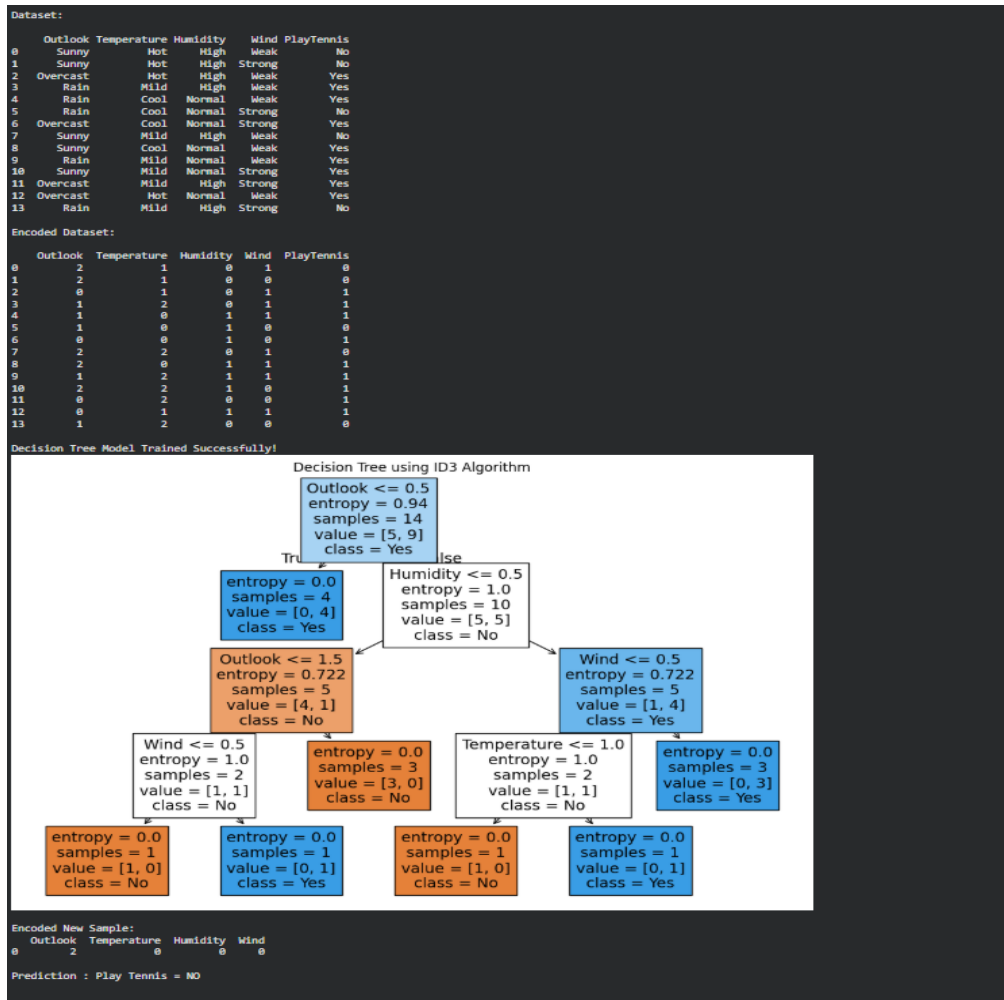
```
encoder = LabelEncoder()
encoder.fit(df[column])
sample_encoded[column] = encoder.transform(sample[column])

prediction = model.predict(sample_encoded)

print("\nEncoded New Sample:")
print(sample_encoded)

if prediction[0] == 1:
    print("\nPrediction : Play Tennis = YES")
else:
    print("\nPrediction : Play Tennis = NO")
```

Output :



Result

The **Decision Tree based ID3 algorithm** was successfully implemented using the Play Tennis dataset. The decision tree was generated based on **Information Gain (Entropy)**, and the trained model correctly classified the given new sample as **Play Tennis = No** (or **Yes**, depending on the input sample). Thus, the experiment demonstrates the working of the **ID3 Decision Tree Algorithm** for classification successfully.